Zhang *et al. Thyroid Research* (2025) 18:48
https://doi.org/10.1186/s13044-025-00268-1

Thyroid Research

REVIEW Open Access

Myxedema coma: challenges and future directions, a systematic survey and review

Yongwen Zhang[1*], Lanfang Chu[2] and Huanhuan Han[1]

Abstract
There was no systematic study on myxedema coma (MC), this review summarizes clinical research on MC conducted over the past 20 years. Publications from 2004 to 2024 were retrieved from databases and included. A total of 584 published cases and 114 unpublished cases of MC were systematically analyzed. The estimated incidence rate of MC was 0.12 (95% confidence interval [CI], 0.10%-0.14%) per million per year. Precipitating factors were identified in 77.6% (95%CI: 73.7%-81.5%) of patients with MC. The overall mortality rate was 38.8% (271/698, 95%CI: 34.9%-42.7%), with shock and multiple organ failure (MOF) being the most common causes of death. The results demonstrated that 88.9% (95%CI: 86.9%-90.9%) of patients with confirmed MC had altered mental status (AMS). Approximately 71.9% (95CI:68.0%-75.8%) of patients with MC had hypothermia, and 66.2% (95CI:62.3%-70.1%) had a heart rate < 60 bpm. Seven signs (hypoglycaemia, hypotension, hypercarbia, hoarseness, dry skin, constipation, and puffiness) showed both negative and positive predictive values below 70% and were consequently excluded from the diagnostic criteria. Combined with literature reviews and statistical analysis, proposed diagnostic criteria for MC are presented. It is generally recommended to confirm the presence of decreased FT3 and/or FT4 levels as a necessary criterion for establishing a diagnosis of MC. Early recognition, clinical suspicion, prompt thyroid hormone replacement, supportive measures, and appropriate management of coexisting problems remain the cornerstone of managing this fatal endocrine emergency. Future research directions for MC should focus on: clinical validation and refinement of diagnostic criteria, elucidating the pathophysiological mechanism underlying MC, and conducting clinical studies to compare the efficacy of different treatment regimens.

Keywords Myxedema coma, Mortality rate, Altered mental status, Thyroid hormone replacement

*Correspondence:
Yongwen Zhang
13914749684@163.com
[1]Department of Endocrinology, Nanjing Integrated Traditional Chinese and Western Medicine Hospital Affiliated with Nanjing University of Chinese Medicine, No.179 Xiaolingwei, Xuanwu District, Nanjing City 210014, China
[2]Department of Gastroenterology, Nanjing Jinling Hospital, General Hospital of Eastern Theatre Command, Nanjing 210012, China

Introduction
Myxedema coma (MC) is an extreme manifestation of untreated hypothyroidism. Most endocrinologists agree that MC represents the most extreme form of hypothyroidism, characterized by altered mental status (AMS), hypothermia, bradycardia, hypoxemia, hyponatremia, hypoglycemia, hypercarbia, renal impairment, hypotension, non-pitting edema, delayed deep tendon reflexes, and the presence of a precipitating factor [1, 2]. Precipitating factors may include infections, cold exposure, stroke, septicemia, metabolic disturbances, cerebrovascular accident, gastrointestinal bleeding, congestive heart

failure, and the use of medications such as analgesic, sedative, hypnotic, antidepressant, antipsychotic, amiodarone, anesthetic and lithium [3–5]. Infections are the primary triggering factor, including urinary tract infections, pneumonia, and cellulitis [6, 7]. MC is a rare disease that can present endocrinologists with a diagnostic challenge, as patients often present with AMS (such as lethargy, confusion, or obtundation) rather than frank coma [8]. Primary hypothyroidism accounts for more than 95% of MC cases, with pituitary or hypothalamic causes accounting for the remaining 5% [2].

A high index of clinical suspicion and awareness of potential precipitating factors for MC are crucial for endocrinologists to facilitate timely diagnosis and treatment. The incidence of MC is also considered to have decreased, owing to improved outpatient detection, increased utilization of health checkups and advancements in the treatment of hypothyroidism. To our knowledge, no systematic review on MC has been published prior to this article. This review summarizes clinical research on MC conducted over the past 20 years. Based on a comprehensive literature review and investigation, recommendations are proposed for the diagnostic criteria and treatment of MC.

Methods

The study included publications from 2004 to 2024 in the databases (National Library of Medicine https://pubmed.ncbi.nlm.nih.gov, China National Knowledge Internet http://www.cnki.net, and pre-printed articles in Research Rquare, MedRxiv, and Arxiv), and using the search terms: "myxedema coma", "severe hypothyroidism", "hypothyroidism AND altered mental status", "subclinical hypothyroidism AND altered mental status", "hypothyroidism AND hypothermia", "subclinical hypothyroidism AND multiple organ failure", "subclinical hypothyroidism AND hypothermia", and "hypothyroidism AND multiple organ failure". Cases with unclear diagnosis, early access, meeting abstract, proceeding paper, letter, and incomplete clinical data were excluded. The study was approved by the ethics committees of Nanjing Integrated Traditional Chinese and Western Medicine Hospital Affiliated with Nanjing University of Chinese Medicine. Continuous variables with normal distribution were analyzed using independent samples *t*-tests. Categorical data were presented as frequencies and percentages (n %), with group comparisons performed using chi-square tests or Fisher's exact test. Logistic regression analysis identified potential risk factors. All statistical tests were two-sided, with *P*-values < 0.05 considered statistically significant. All analyses were conducted using SPSS version 22.0 (IBM Corp).

Results

Definition of MC

MC or myxedema crisis represents the ultimate stage of severe hypothyroidism. It is characterized by extremely low thyroid hormone levels and carries a high mortality rate, often proving fatal despite treatment [3–9]. MC is a rare but still recognized end-stage manifestation of long-term hypothyroidism [10], affecting most organs systems and potentially leading to MOF in severe cases. This condition frequently presents in winter and is associated with symptoms of hypothyroidism. A recognized feature of MC is the observed discrepancy between clinical presentation and thyroid function test results [11]. The characteristic clinical manifestations reflect decompensated hypothyroidism with impaired thermoregulation and AMS [3–12]. Furthermore, thyroid hormone levels in MC show poor correlation with disease prognosis and severity [13].

Epidemiology of MC

According to data from Western countries, the incidence of MC is approximately 0.22 per million per year [8]. A systematic analysis of 584 published cases and 114 unpublished MC cases (including four cases treated by the author) yielded an estimated incidence rate of 0.12 (95% confidence interval [CI], 0.10%-0.14%) per million per year. This represents 0.00016% of all hypothyroid patients and 0.0034% of hospitalized hypothyroid patients. Management of MC cases by endocrinology units has gradually increased in recent years (2012–2024), while emergency department cases have decreased (Fig. 1). This shift may reflect heightened physician awareness of MC, earlier detection through widespread physical examination, and discussion of cases at thyroid conferences. Precipitating factors were identified in 77.6% (95%CI: 73.7%-81.5%) of MC patients, with infection being the most common trigger, followed by cold exposure. Mortality among confirmed or suspected MC cases was 38.8% (271/698, 95%CI: 34.9%-42.7%), primarily due to shock and MOF (Fig. 2). Simple regression analysis identified cardiac complications (odds ratio [OR] = 10.060, 95%CI: 6.421–13.699, $p < 0.001$), Glasgow Coma Scale (GCS; OR = 0.846, 95%CI: 0.421–1.271, $p = 0.0062$), and advanced age (OR = 2.582, 95%CI: 1.056–4.108, $p = 0.015$) as independent prognostic factors for mortality. However, after adjusted for risk factors (gender, age, TSH levels, insurance type, coronary artery disease, chronic kidney disease, diabetes mellitus, hypertension, hyperlipidemia, smoking status, alcohol intake, and BMI), multiple logistic regression demonstrated that the high-dose levothyroxine (T4) therapy (OR = 2.414, 95%CI: 1.145–3.683, $p = 0.017$), comorbidities of shock (OR = 2.534, 95%CI:1.422–3.646, $p = 0.019$), and MOF (OR = 14.638; 95%CI: 5.894–23.382, $p < 0.001$) were

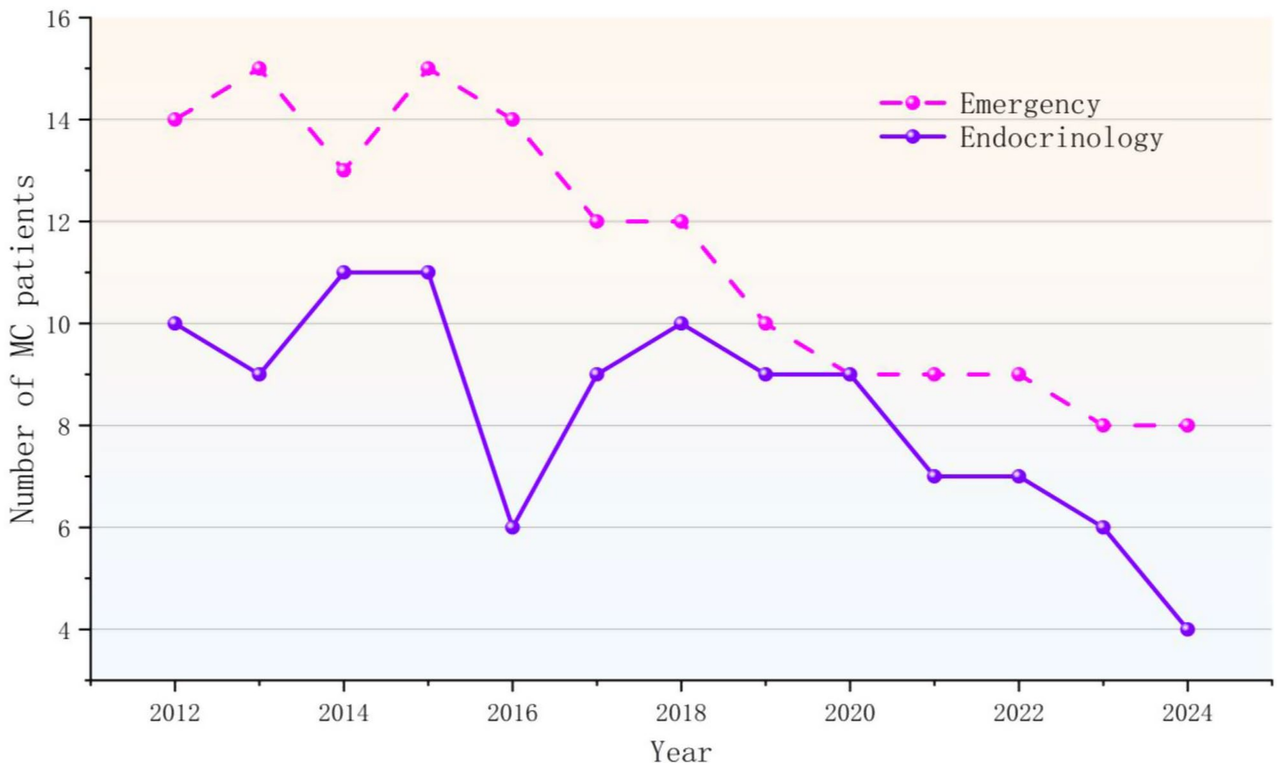

Fig. 1 MC patient distribution changes: Emergency vs. Endocrinology ($n = 256$)

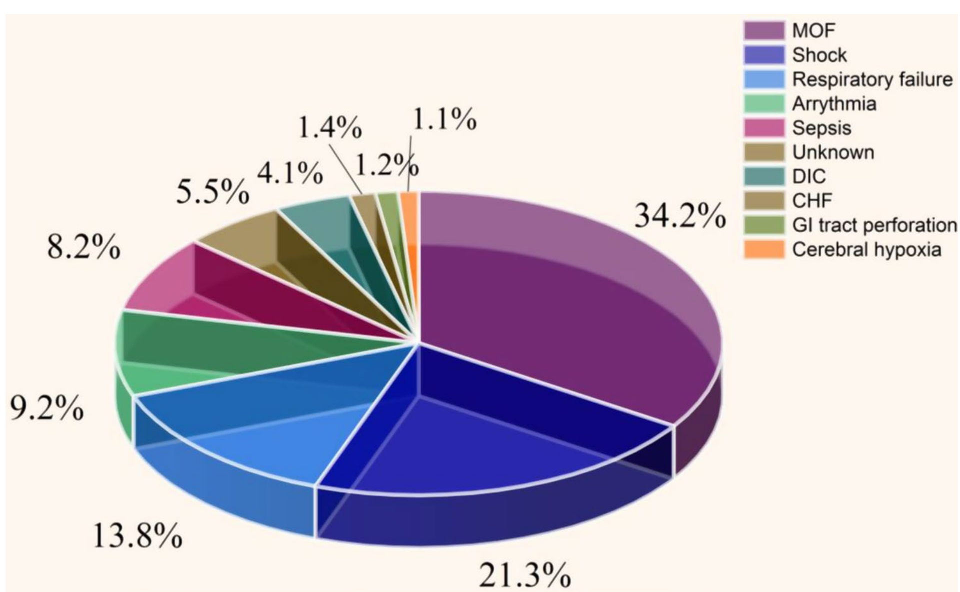

Fig. 2 MC mortality causes. MOF: multiple organ failure; GI: gastrointestinal; DIC: disseminated intravascular coagulation; CHF: congestive heart failure

significantly associated with mortality. The mean age of MC patients was 52.1 ± 12.1 years (range 49–83 years) in this review, showing no significant difference from general hypothyroidism population. However, mortality significantly differed between patients aged >65 years and those aged < 65 years.

Pathophysiology of MC

Although the pathogenesis of MC remains incompletely understood, low triiodothyronine (T3) levels secondary to hypothyroidism constitute its primary cause [7]. This deficiency may result from T3 transporter deficiency, central stimulation defect, or deiodinase deficiency affecting thyroid hormone transformation [7, 14, 15]. Notably, destruction of over 90% of the thyroid cells typically precedes hypothyroidism onset [16, 17]. During MC, hypothalamic regulation probably suppresses compensatory mechanisms, leading to a near-undetectable decline in serum T3 levels. Concurrently, markedly reduced serum T4 concentrations are strongly associated with high mortality risk [9]. Understanding the mechanisms underlying MC's clinical manifestations requires recognition that measured thyroid-stimulating hormone (TSH) or free thyroxine (FT4) levels alone are insufficient to explain its signs and symptoms. This complexity arises because thyroid hormone's tissue level effects involve genomic and postgenomic mechanisms. The most common precipitating factors for MC are infection and inflammation-induced cytokines such as interleukin-1 and tumor necrosis factor alpha (TNF-a). These cytokines regulate the expression of numerous proteins involved in thyroid hormone metabolism [18], potentially activating inflammatory signaling pathways like nuclear factor kappa-light-chain-enhancer of activated B cells (NFjB) and activated protein 1. Such activation may alter enzymes related to thyroid hormone transporters, metabolism, and receptors. Consequently, infections are strongly implicated in disrupting thyroid hormones metabolism and intracellular levels of multiple systems. While the pathogenesis involves additional complexity, other precipitating factors including metabolic disturbances and cold exposure are likely to exert similar effects on the actions and metabolism of the thyroid hormone [2].

Clinical manifestations of MC

Symptoms and signs are typically exacerbations of the hypothyroidism and may include AMS [19]. Based on the clinical symptoms and signs, MC should be readily recognized, but it is frequently missed clinically. Following brain stem infarction, elderly patients presenting with symptoms and signs of the hypothyroidism may concurrently develop hypothermia and coma.

The study demonstrated that 88.9% (95%CI: 86.9%-90.9%) of patients with definite MC manifested AMS. Hypothermia was present in 71.9% (95CI:68.0%-75.8%) of MC patients. Bradycardia (heart rate < 60 bpm) was observed in 66.2% (95CI:62.3%-70.1%) of definite MC patients. Hypoxemia affected 59.4% (95CI:55.5%-63.3%) of MC patients, while severe hypoventilation was found in 52.9% (95CI:49.0%-56.8%) of definite MC patients and 58.9% (95CI:55.0%-62.8%) of suspected MC patients. Hyponatremia was present in 58.3% (95CI:54.4%-62.2%) of MC patients (Fig. 3). Among the definite MC patients, 70.3% (95CI:66.4%-74.2%) exhibited more than three signs or manifestations. The most common combination (12.1% [95CI:10.1%-14.1%]) of definite MC patients was concomitant presence of AMS, hypothermia and bradycardia, along with precipitating factors. The second most common combination (8.6% [95CI:6.6%-10.6%]) of definite MC patients was AMS, hypothermia, and bradycardia (Fig. 4).

Central nervous systems

Intellectual functions universally slow in the absence of thyroid hormone deficiency [20]. In MC patients, there may be a history of poor memory, lethargy, cognitive impairment, depression, or even psychosis. Positron

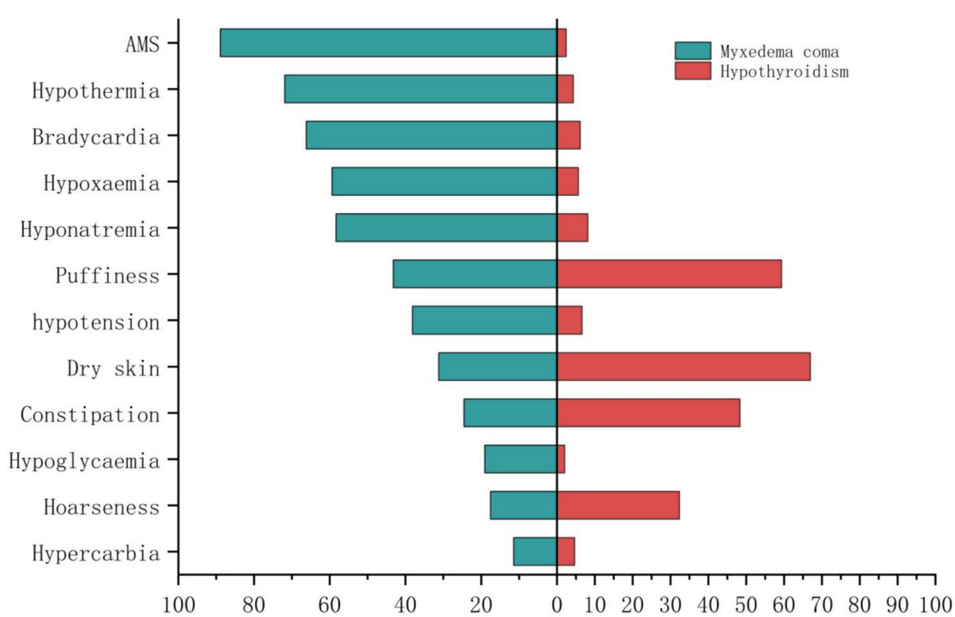

Fig. 3 Frequency of symptoms and signs (%) in patients with MC ($n = 656$) and hypothyroidism ($n = 667$). AMS: altered mental status

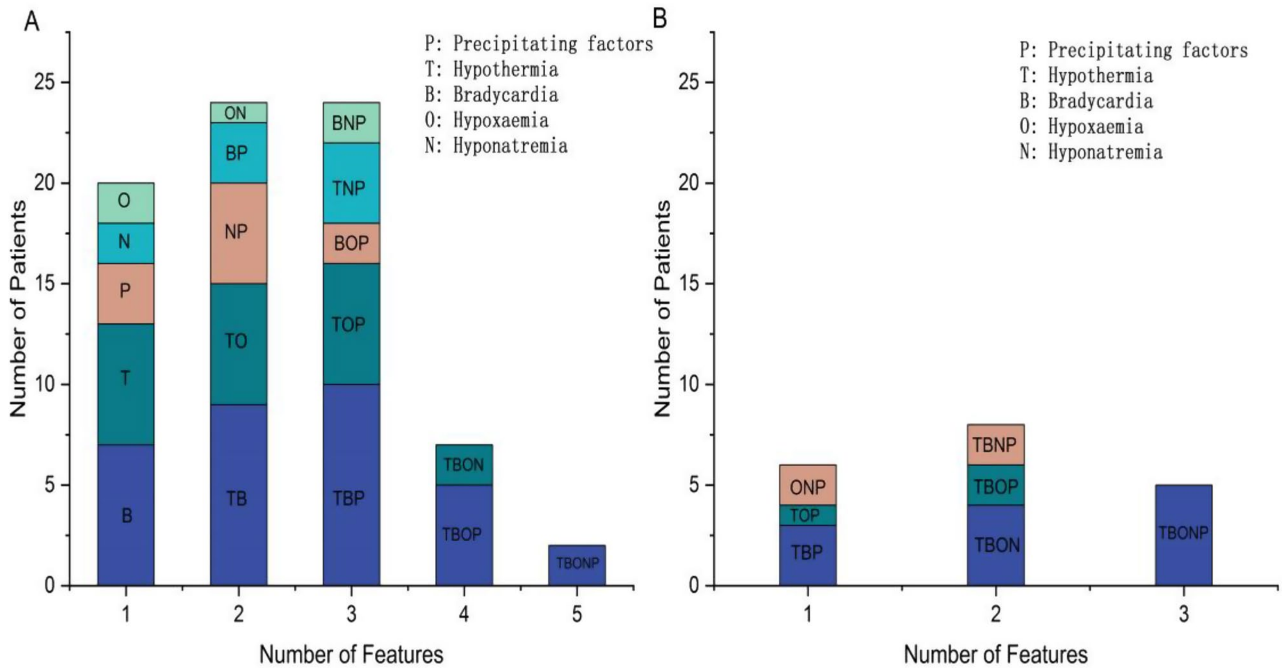

Fig. 4 Manifestation patterns in MC patients. (**A**) With AMS features ($n = 228$); (**B**) Without AMS features ($n = 28$)

emission tomography (PET) brain scans before and after thyroid hormone replacement revealed that reversibly reduction in glucose uptake within specific brain areas [21]. Tendon reflexes demonstrate slowing and produce characteristic "hung-up reflexes". Histopathologic brain examination indicates nervous system edema with mucinous deposits in nerve fibers.

Hypothermia

Literature survey reveals that most cases with MC develop in winter, suggesting cold weather may lower the susceptibility threshold. In fact, hypothermia is present in nearly all MC patients, concurrent infection can mask it, resulting in normal body temperature. Angel and Sash emphasized that hypothermia itself warrants immediate thyroid hormone replacement therapy [22]. Persistent hypothermia is a poor prognosis sigh, and low ambient temperatures constitute a risk factor. It has been confirmed that the ultimate response to treatment is related to the degree of body temperature.

Cardiovascular manifestations

Cardiovascular manifestations in MC are numerous, including bradycardia, decreased myocardial contractility, cardiomegaly, prolonged QT interval, non-pitting edema and hypotension [23]. In MC, the cardiac silhouette is typically enlarged. Reduced myocardial contractility leads to low cardiac output and stroke volume. Hypotension may be develop due to cardiovascular collapse and reduced intravascular volume. Shock can occur in advanced stage of the MC [24]. During shock, vasopressor therapy often proves ineffective unless T4 is given at the same time. Appropriate thyroid hormone replacement therapy can restore cardiovascular function.

Respiratory system

Myxedema of the respiratory muscles, combined with inhibition of the hypercarbonic ventilation and hypoxic drives, promotes carbon dioxide retention and alveolar hypoventilation, which contribute to MC pathogenesis. Respiratory function is further compromised by reduced lung volumes. Pleural effusions, macroglossia, and edema of the larynx and nasopharynx reduce effective airway opening. Decreased hypoxic respiratory and driven ventilation may be the cause of respiratory depression in MC; however, obesity and respiratory muscle dysfunction may aggravate the insufficiency of ventilation [9].

Gastrointestinal manifestations

Patients with MC commonly present with abdominal pain, nausea, anorexia, and constipation with fecal retention. Abdominal distention accompanied by colic and vomiting may mimic mechanical intestinal obstruction [25]. Abdominal dilation, megacolon, or paralytic ileus may occur, and decreased intestinal motility is frequently observed. Neurogenic oropharyngeal dysphagia is associated with delayed swallowing and aspiration pneumonia [26]. Gastric atony may reduce the absorption of oral T4. Histologic examination reveals myxedematous infiltration of the intestinal wall with mucosal atrophy affecting both intestinal and gastric tissues.

Renal and electrolyte manifestations

Urine flow is reduced, the decreased of urine volume in distal dilution segment of nephron lead to the delay in water excretion [27]. Low glomerular filtration rate and hyponatremia consistently manifest in MC patients. The hyponatremia is caused by inability to discharge the water load, which is due to excess vasopressin secretion and reduced water delivery to the distal nephron. Diuretic use may exacerbate hyponatremia while masking the edema associated with myxedema. Thyroid hormone replacement reverses these pathophysiological changes [9].

Diagnostic challenges

The diagnosis of MC is challenging due to its insidious onset and the rarity of the condition. The absence of established diagnostic criteria further impedes early recognition. Serum T3 and T4 levels are decreased, accompanied by either elevated TSH in patients with primary hypothyroidism or low/normal TSH in patients with secondary hypothyroidism. The diagnosis of MC relies on clinical history, physical examination, and exclusion of other causes of coma, it should be considered in any patient with AMS or coma. Crucially, it should be emphasized that MC patients rarely exhibit all classic manifestations. There were no published diagnostic criteria for MC. Given MC's rarity, prospective studies to develop diagnostic criteria are considered time consuming and very difficult [28]. During criteria development, consensus established hypothyroidism as an absolute diagnostic requirement. Subsequently, the prevalence of clinical features was compared between MC patients and those with hypothyroidism but not MC. Concurrently, the combination patterns of clinical manifestations were analyzed (Figs. 3 and 4). Due to the specificity of AMS for MC, the absence or presence of AMS was used to stratify the patterns (Fig. 4).

Table 1 Diagnostic accuracy metrics for 12 MC features

Symptoms and signs	Sensitivity (%)	Specificity (%)	Positive predictive value (%)	Negative predictive value (%)
AMS	96.5	50.0	94.5	80.0
Hypothermia	91.1	39.1	88.9	45.0
Bradycardia	92.4	39.3	86.5	55.0
Hypoxaemia	36.0	98.7	96.5	60.7
Hyponatremia	60.0	96.3	94.2	70.7
Hypoglycaemia [a]	58.4	31.0	42.3	45.8
Hypotension [a]	64.9	26.0	41.3	48.1
Hypercarbia [a]	63.6	57.8	60.1	62.7
Hoarseness [a]	65.3	31.0	34.8	61.4
Dry skin [a]	55.3	39.6	39.6	55.3
Constipation [a]	52.9	30.0	46.2	36.0
Puffiness [a]	78.1	50.8	62.5	68.9

[a] Seven signs (hypoglycaemia, hypotension, hypercarbia, hoarseness, dry skin, constipation, and puffiness) showed PPV and NPV were below 70%, therefore, excluded from the diagnostic criteria

Following comprehensive analysis, the recommended diagnostic criteria for MC were established based on clinical characteristics and the clinical course of these patients. Seven signs (hypoglycaemia, hypotension, hypercarbia, hoarseness, dry skin, constipation, and puffiness) demonstrated the values of negative and positive predictive below 70% and were consequently excluded from the diagnostic criteria (Tables 1 and 2; Fig. 5). Combined with literature reviews and statistical analysis, proposed diagnostic criteria for MC are presented in Table 2. It is generally recommended to confirm the presence of decreased FT3 and/or FT4 levels as a necessary criterion for establishing a diagnosis of MC. However, it is well known that patients with severe non-thyroid disease frequently exhibit low levels of thyroid hormones, particularly T3. As a matter of fact, some confirmed MC cases presented with normal levels of FT3 and FT4 and elevated levels of TSH. A significant diagnostic pitfall is overlooking MC in patients with subclinical hypothyroidism. Endocrinologists should be wary of the possibility of MC even in patients with subclinical hypothyroidism [29]. Therefore, the recommended diagnostic criteria for MC should include reduced serum FT4 and/or FT3 levels, accompanied by either an elevated TSH in primary hypothyroidism or a low/normal TSH in secondary hypothyroidism, as an essential component. Despite high sensitivity (100%) and specificity (85.71%) of the published MC diagnostic scoring system [30], the proposed diagnostic criteria demonstrated significantly higher accuracy and positive predictive value (PPV) (χ^2=8.554, 12.766; p < 0.001), indicating superior performance for these metrics. Therefore, it is necessary to expand the diagnostic criteria for MC, but there is a need for further studies.

Severity and prognosis of MC

MC represents the most extreme form of hypothyroidism and is associated with high mortality rates even under optimal conditions and treatment. Early recognition and therapy are essential, unrecognized and untreated, the MC is usually fatal. Although the mortality rate of MC has decreased in recent years, mortality is still high, ranging from 25% to 60%, even with prompt treatment [4]. Maintaining a high degree of suspicion is crucial, particularly during cold winter months when encountering an elderly woman with symptoms and signs compatible with hypothyroidism. This presentation emphasizes careful evaluation for patients with disparate thyroid hormone levels.

In this study, the most common causes of death, such as MOF and shock, were largely consistent between the confirmed and suspected groups. This finding indicates

Table 2 Definition and proposed diagnostic criteria for MC

Definition of MC

MC or myxedema crisis represents the ultimate stage of severe hypothyroidism. It is characterized by extremely low thyroid hormone levels and carries a high mortality rate, often proving fatal despite treatment. MC is a rare but still recognized end-stage manifestation of long-term hypothyroidism, affecting most organs systems and potentially leading to multiple organ failure in severe cases.

Prerequisite for diagnosis:

Serum FT4 and/or FT3 are reduced with either increase of TSH in primary hypothyroidism or low/normal TSH in secondary hypothyroidism.

Symptoms:

1. Altered mental status (AMS) manifestations: confusion, lethargy, slowed mentation, poor memory, cognitive dysfunction, depression, psychosis, dementia, or coma (Glasgow Coma Scale ≤ 14).
2. Hypothermia: ≤95°F(35℃).
3. Bradycardia: ≤60 beats per minute.
4. Hypoxaemia.
5. Hyponatremia.
6. Precipitating factors.

Proposed Diagnostic Criteria

Grade of MC	Diagnostic feature combinations
Definite MC	First **diagnostic criteria** Hypothyroidism plus at least one AMS manifestation and one of the following: hypothermia, bradycardia, hypoxaemia, hyponatremia, or precipitating factors. Alternate **diagnostic criteria** Hypothyroidism and at least three of the following: hypothermia, bradycardia, hypoxaemia, hyponatremia, or precipitating factors.
Suspected MC	First **diagnostic criteria** Hypothyroidism and combination of two of the following: hypothermia, bradycardia, hypoxaemia, hyponatremia, or precipitating factors. Alternate **diagnostic criteria** Fulfills definitive MC diagnostic criteria, except for unavailable serum FT3/FT4 levels.

Exclusion

Patients are excluded if diagnosed with any of the following diseases: impaired consciousness (e.g., psychiatric disorders and cerebrovascular disease), exposure to cold, drug intoxication, or electrolyte disturbance.

Provisions

When it is difficult to determine whether a symptom is caused by MC or solely by an underlying condition, attribute the symptom to MC triggered by those factors.

that mortality of patients with MC depends on the severity of complications. Univariate regression analysis revealed a consistent association between AMS and mortality. However, these were not significant independent factors for high mortality rate of MC in multiple regression analysis. This may be because the latest advances in the management of critically disease reduced mortality associated with these factors. The study has indicated that mortality for MC was 38.8% (271/698, 95%CI: 34.9%-42.7%), indicating a continued need for improved prognosis of patients with MC. The study attempts to determine the disease severity, or the perceived severity of patients with MC, based on the following two indexes: the Sequential Organ Failure Assessment (SOFA) score and Acute Physiology and Chronic Health Evaluation (APACHE) II score [31, 32]. In this study, the means ± SD for SOFA and APACHE II scores were 3.41 ± 0.35 (range 0–12) and12.32 ± 0.51 (range 0–37), respectively. The scores of death patients were significantly higher than that of survival patients: died versus survived = 4.30 ± 1.42 versus 3.00 ± 1.05 (p = 0.032) for SOFA and 15.21 ± 4.02 versus 10.85 ± 4.22 (p = 0.015) for APACHE II, respectively.

Discussion

Management and treatment challenges

Treatment of MC should be initiated as rapidly as possible, given its high mortality rate and can be started even before confirmation of serum thyroid hormone levels. Because MOF is the characteristic of MC, management requires a multidisciplinary approach, including endocrinologists, neurologists, and cardiologist. Based on the literature reviews, suggestions have been established for the management of MC. These suggestions include: thyroid hormone replacement; appropriate management of coexisting conditions such as infection; supportive measures; ventilatory support; correction of abnormal electrolyte levels and hypotension; management of precipitating factors, and steroid therapy. Owing to the high mortality, all patients with MC must be admitted to an intensive care unit (ICU).

Thyroid hormone replacement

Thyroid hormone therapy remains the cornerstone of MC treatment. Despite its critical importance for patient survival, controversies persist regarding the optimal dosage, route of administration, and frequency of treatment. Some experts favor administration of T4, some of T3, while others advocate for combination therapy. Most authorities recommend intravenous T4 administration, primarily because gastrointestinal absorption may be impaired [33]. An alternate method is administration through nasogastric tube; however, absorption is uncertain and there is a risk of aspiration, particularly in

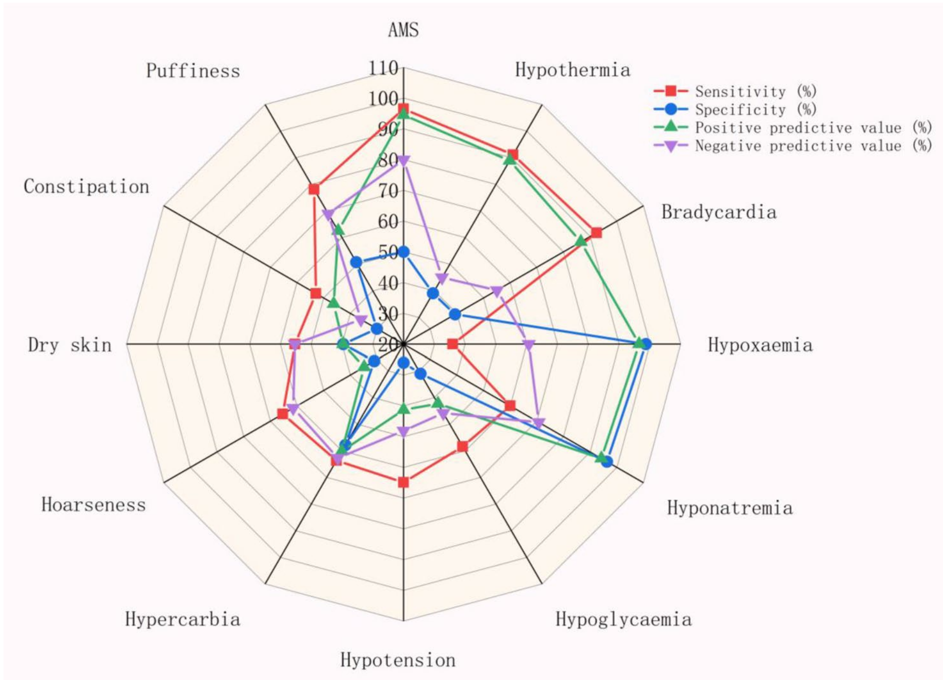

Fig. 5 Radar chart: Diagnostic accuracy of 12 MC indicators

the presence of gastric atony. It is worth noting that T4 must be administered by direct syringe injection and not through an infusion tube, as up to 40% of the initial concentration may be lost due to adherence to the polypropylene tube [34].

T4 treatment provides a slow and smooth onset of action and is associated with few adverse effects. To address the organism's need for T3, considering the physiologic monodeiodination of T4 and its conversion to T3, many endocrinologists prefer supplementing the peripheral hormone pool to T4 with an initial intravenous dose of 100–500 µg.This approach may yield clinical improvements within hours, although significant effects typically occur after 24 h. Subsequently, daily intravenous doses of T4 is 50 to 100 µg are administered [2, 3, 9]. Increasing the initial dose leads to a rapid rise in serum T4 concentrations, often exceeding the normal range, followed by a decrease to the normal reference value within 24 h. Concurrently, serum T3 concentrations begin to increase, and serum TSH concentrations start to decrease as T4 is converted to T3. However, in severe cases of MC, conversion of T4 to T3 may be reduced, leading some endocrinologists to advocate for the use of T3 in the treatment of MC. It is important to note that thyroid hormone replacement carries a risk of precipitating angina, heart failure and arrhythmia. Therefore, treatment should be initiated cautiously, particularly in elderly patients and those with underlying cardiovascular disease.

The T3 therapy carries the risk of worsening or precipitating cardiac complications, which conflicts with the concern that untreated MC might be fatal. Endocrinologists who support T3 therapy believe that the onset of T3 (1/2 to 3 h) is faster than T4 (>24 h), potentially increasing the patient's chance of survival. With the successful experience reported by Barbetakis and colleagues, many endocrinologists adopted the combination therapy may improve the clinical outcomes and shorten the response time. Studies conducted in baboons suggest that T3 crosses the blood-brain more readily than T4, thereby potentially affecting neuropsychiatric symptoms. Given the high mortality of MC, many experts advocate for rapidly increasing thyroid hormones levels. Accordingly, many clinical thyroid specialists administer an initial intravenous dose of 200 to 300 µg T4 and 25 µg T3, then use 100 µg T4 and 25 µg T3 after 24 h, followed by 50 µg T4 daily until the patient regains consciousness [9]. Once the patient tolerates oral intake, oral T4 should be administered.

No guidelines can take into account all relevant factors, such as age, comorbid conditions, intrinsic cardiovascular function, and neuropsychiatric status, which may influence replacement dosing due to abnormalities in drug metabolism and distribution. The study demonstrated that mortality in MC is associated with high-dose T4 therapy and advanced age. However, a 500 µg T4 loading dose was found to be safe in younger patients, whereas lower doses are recommended for elder patients with MC. Determining optimal therapy is challenging due to the absence of clinical studies comparing the efficacy of different treatment regimens.

Supportive measures and management of coexisting problems

Thyroid hormone replacement may require several days to improve patient's clinical condition. Consequently, supportive care is crucial for optimizing the survival of patients [35]. These include correction of hypothermia, mechanical ventilation, monitoring and treatment in ICU, maintenance of fluid balance, treating and avoiding potential precipitants, using warming blankets, and managing any coexisting conditions.

Antibiotic therapy

Infection is a common symptom of MC, but the symptoms of infection may be masked, the patient with MC may not have fever despite significant infection. In fatal cases, sources of infection are frequently unrecognized [35]. Broad-spectrum antibiotics are therefore essential, initial antibiotic should be empirical, with timely adjustment of the regimen if symptoms of pneumonia or other specific infections develop.

Ventilatory support

To ensure sufficient oxygenation, insertion of an endotracheal tube may be necessary. Most patients with MC require mechanical assisted ventilation, and antibiotic therapy should be considered regardless of the cause of respiratory depression. Mechanical ventilatory support is particularly required in MC patients whose coma is caused by certain medications. Due to the potential delayed recovery, some patients may need ventilation for several weeks, even after initiating of T4 therapy.

Hypothermia

In hypothermic MC patients, passive rewarming is required. Undergo procedures in a warm ICU is beneficial in patients with MC, as core temperature can be continuously monitored. Electric blankets are recommended for active warming but should be used cautiously, because decreased peripheral vascular resistance may lead to the risk of hypotension. Thyroid hormone replacement is essential to restore normal body temperature, but hypothermia may persist for several days despite treatment initiation.

Hypotension

Because external warming may aggravate hypotension, therefore, intravenous volume repletion should be initiated with caution, glucose supplementation and hypertonic saline may be required to alleviate the potential hypoglycemia and severe dilutional hyponatremia. Vasopressors are often necessary in MC patients to maintain blood pressure until T4 therapy takes action. Hydrocortisone is routinely administered, particularly if primary adrenal insufficiency or pituitary disease is suspected, because it has the effect of stabilizing vessels.

Glucocorticoid therapy

Primary adrenal insufficiency may coexist in patients with autoimmune primary hypothyroidism due to Hashimoto's disease. Thyroid hormone replacement may increase cortisol clearance and exacerbate cortisol deficiency, partly attributable to apoptosis or degeneration of the adrenal glands, which reduces hormone production capacity [36]. Thus, hydrocortisone administration is necessary before initiating thyroid replacement therapy [37]. Other laboratory and clinical features may be found in patients with MC complicated with adrenal insufficiency, such as hyponatremia, hypotension, hypercalcemia, lymphocytosis, hyperkalemia, and azotemia. To prevent acute adrenal insufficiency, a common complication in patients with MC, hydrocortisone is typically administered intravenously (100 mg every 8 h). Subsequently, based on further diagnostic evaluation and clinical response, hydrocortisone can be considered to reduce and discontinue. This short-term glucocorticoid therapy is safe and can be discontinued when the pituitary-adrenal function is confirmed adequate and patient's condition stabilizes.

Future directions

The study has several advantages, including a comprehensive scope, clinical utility for decision making, and a large dataset. The limitations of the study include a relatively small cohort size, diagnostic criteria derived primarily from statistical analyses, and a lack of clinical validation. To improve the outcome of patients with MC in the absence of local guidelines, the adoption of the management strategies discussed in this study is recommended. Implementation of these protocols may help achieve improved outcomes in patients with MC. Importantly, the pathogenesis leading to the development of MC has not been fully understood. Further investigation into its mechanisms is essential, because this will definitely help improve the understanding of the disease process and reduced mortality of MC. Future study direction of MC includes clinical validation and improvement of diagnostic criteria for MC, pathophysiological mechanism of MC, and clinical studies comparing the efficacy of different treatment regimens. In order to overcome these diagnostic and management challenges, the collaborative efforts of governmental bodies, policy makers and health institutions are needed to ensure the accessibility and availability of the necessary resources for optimal treatment in MC.

Conclusion

Early recognition, clinical suspicion, prompt thyroid hormone replacement, supportive measures, and appropriate management of coexisting conditions remain central to treating this fatal endocrine emergency. The lack of established diagnostic criteria hinders early recognition of MC. Given its high mortality rate, the treatment of MC should be initiated as rapidly as possible. Because thyroid hormone replacement may take several days to improve the clinical condition of patients, while awaiting clinical improvement, supportive care is critically important during this period.

Acknowledgements
None.

Author contributions
YWZ designed the study protocol and drafted the manuscript. HHH and LFC checked the manuscript. All authors contributed to drafting the manuscript and have read and approved the final manuscript.

Funding
No financial support was granted for this research by funding entities in the public, commercial, or non-profit sectors.

Data availability
The data supporting the findings of this study are available from the corresponding authors upon reasonable request.

Declarations

Competing interests
The authors declare no competing interests.

Received: 14 May 2025 / Accepted: 22 September 2025
Published online: 07 October 2025

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